

EAST WEST UNIVERSITY

Department of Mathematical and Physical Sciences

Course Outline

Semester: Spring 2025

Course Title	: Differential Equations and Special Functions
Course Code	: MAT102
Pre-Requisite	: MAT101
Credit Hour	: 3
Course Instructor	: Md. Ashraf Hossain, Lecturer, Adjunct Faculty
Office Room	: FUB (9 th floor, Adjunct Faculty Room)
E-mail Address	: ashraf.hossain@ewubd.edu
Text Book	: A First Course in Differential Equations with Modeling Applications, (Tenth Edition) Dennis G. Zill, Publisher: Brooks/Cole Cengage.
Reference Books	: (a) Differential Equations by Shepley L. Ross (Third Edition). (b) Ordinary & Partial Differential Equations by M D Raisinghania. (c) Advanced Engineering Mathematics by H.K Dass.
Class Hours	:

Section	Days	Time	Room
18	TR	08:30 am-10:00 am	FUB-802

Course Goal

This course is designed to develop the basic concept of students in the area of ODE and PDEs. A student has to study this course because the contents of the course are very much applicable in all branches of Science and Engineering.

Course contents:	1. Ordinary Differential Equations. 2. Partial Differential Equations. 3. Special Functions.
Description:	The following topics will be covered throughout the semester. Ordinary Differential Equations: 1. Degree and order of ordinary differential equations. Formations of differential equations. 2. Solutions of first order differential equations; Separable & homogeneous equations, exact equation. Integrating factor. Equations made exact by integrating factors. First order linear equation. Bernoulli's equation. 3. Higher order linear homogeneous equation with constant coefficients. Initial and boundary value problems. Linear non-homogeneous equation with constant coefficients: Method of undetermined coefficients, Method of variation of parameters. Operator method. Series solution; Frobenius method. Partial Differential Equations: 1. Formation of PDEs & First order linear PDEs. Solution of PDEs of first order; Lagrange's method. 2. Second order homogeneous & non-homogeneous PDEs with constant coefficients. 3. Wave equations. Particular solutions with boundary and initial conditions. Special Functions: 1. Legendre differential equation and Legendre polynomials, Recurrence relations for Legendre polynomials, Spherical harmonics. 2. Bessel differential equation, Bessel functions, Recurrence relations for Bessel functions, Modified Bessel functions. 3. Hermite differential equation, Hermite polynomials, Hyper-geometric function.

Assessment Tools

Assessment tools include **Class Tests (Short Quizzes), Assignments, Presentations/Viva, and Exams**. The Class Tests are about 10-15 minutes in class and the Mid Term Exam and Final Exam are 80 minutes duration in class. Class Test dates will be announced in the Class and the Exam dates are as follows:

Mid Term Exam Date	08/04/2025	
Final Exam Date	25/05/2025	

Assessment/Evaluation/Grading Policy

The relative contributions of class tests, presentations and exams are as follows:

Test/Exam	% of Marks
Class Tests	20
Assignments	10
Presentation/Viva	10
Mid Term Exam	30
Final Exam	30

The University Grading Scheme is the following

Range of Marks (%)	Letter Grade	Grade Point	Range of Marks (%)	Letter Grade	Grade Point	Range of Marks (%)	Letter Grade	Grade Point
80 – 100	A+	4.00	65 – 69	B+	3.25	50 – 54	C+	2.50
75 – 79	A	3.75	60 – 64	B	3.00	45 – 49	C	2.25
70 – 74	A-	3.50	55 – 59	B-	2.75	40 – 44	D	2.00
						Less than 40	F	0.00

Special Instructions :

- ♣ No make-up quizzes will be held.
- ♣ Students are requested to **switch off their mobile telephone** during the class hour.
- ♣ There is **zero tolerance for cheating** at EWU. Students caught with cheat sheets in their possession, whether used or not used, &/or copying from cheat sheets, writings on the palm of hand, back of calculators, chairs or nearby walls, etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.

Md. Ashraf Hossain

Date: 11-2-2025